TITLE: Predictive Modeling of Academic Performance Using Behavioral Analytics

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Abstract

*Early identification of students at risk of poor academic performance is a critical challenge in modern education systems. This study presents **EduPredict AI**, a machine learning–based framework designed to forecast student academic success using behavioral analytics derived from Learning Management System (LMS) interactions. The proposed model employs a Random Forest classifier optimized through GridSearchCV. Experimental results demonstrate a testing accuracy of **86.46%**, significantly outperforming baseline approaches. Feature importance analysis reveals that behavioral indicators such as resource engagement and class participation contribute approximately 59% more predictive importance than demographic features. These findings highlight the potential of data-driven approaches to support early and targeted academic intervention strategies.*

1. Introduction

Educational institutions increasingly rely on digital platforms to support teaching and learning. These platforms generate large volumes of student interaction data, often referred to as digital footprints.

Traditional grading systems typically identify struggling students only after academic performance has declined. Such reactive approaches limit the ability of educators to provide timely and effective interventions.

This study proposes a proactive predictive framework that analyzes behavioral engagement data extracted from LMS systems to identify potential academic risks early.

The main objectives of this study are:

- To develop a machine learning model capable of predicting student academic performance.
- To identify the most influential behavioral features affecting academic success.
- To provide interpretable insights that educators can use for targeted interventions.

This study contributes to the field of educational data mining by demonstrating the effectiveness of behavioral analytics combined with explainable artificial intelligence in predicting academic performance.

2. Related Work

Numerous studies have explored the use of machine learning techniques in educational data mining.

Algorithms such as:

- Random Forest
- Support Vector Machines
- Logistic Regression
- Neural Networks

have been widely applied to predict academic performance.

Many existing studies rely heavily on demographic variables such as gender, socioeconomic status, or parental education. However, recent research indicates that behavioral engagement metrics extracted from LMS platforms may provide stronger predictive signals.

In contrast, EduPredict AI focuses primarily on student engagement indicators, including participation frequency and resource interaction, to build a more actionable predictive model.

3. Dataset Description

This study utilizes the ***xAPI-Edu-Data*** dataset, which contains records of students' behavioral and demographic data collected from a Learning Management System (LMS) environment. The dataset is licensed under Creative Commons Attribution 4.0 International License.

The dataset contains 480 students with 16 features, including key behavioral indicators such as:

Table 1. Dataset Features and their Description

Feature	**Description**
raisedhands	Number of times a student participates in class
VisITedResources	Number of learning materials accessed
AnnouncementsView	Number of announcements viewed
Discussion	Participation in discussion forums
StudentAbsenceDays	Frequency of student absences

The target variable represents student academic performance levels, categorized into:

- Low Performance
- Medium Performance
- High Performance

4. Methodology

The development of the EduPredict AI framework followed a structured machine learning pipeline, including data preprocessing, feature engineering, model training, and evaluation.

4.1 Data Preprocessing

Data preprocessing is a critical step in ensuring the quality and reliability of the predictive model.

The dataset initially contained 480 student records but to maintain data integrity, records containing duplicate values were removed, resulting in a cleaned dataset of 478 complete observations.

Categorical variables were transformed into numerical representations using One-Hot Encoding, implemented through the ***pandas.get_dummies() function.*** This transformation enables machine learning algorithms to effectively process non-numeric attributes by converting them into binary indicator variables.

Additionally, all features were reviewed to ensure consistency and relevance for modeling.

4.2 Feature Engineering and Selection

The original dataset consisted of 16 features, including both behavioral and demographic attributes.

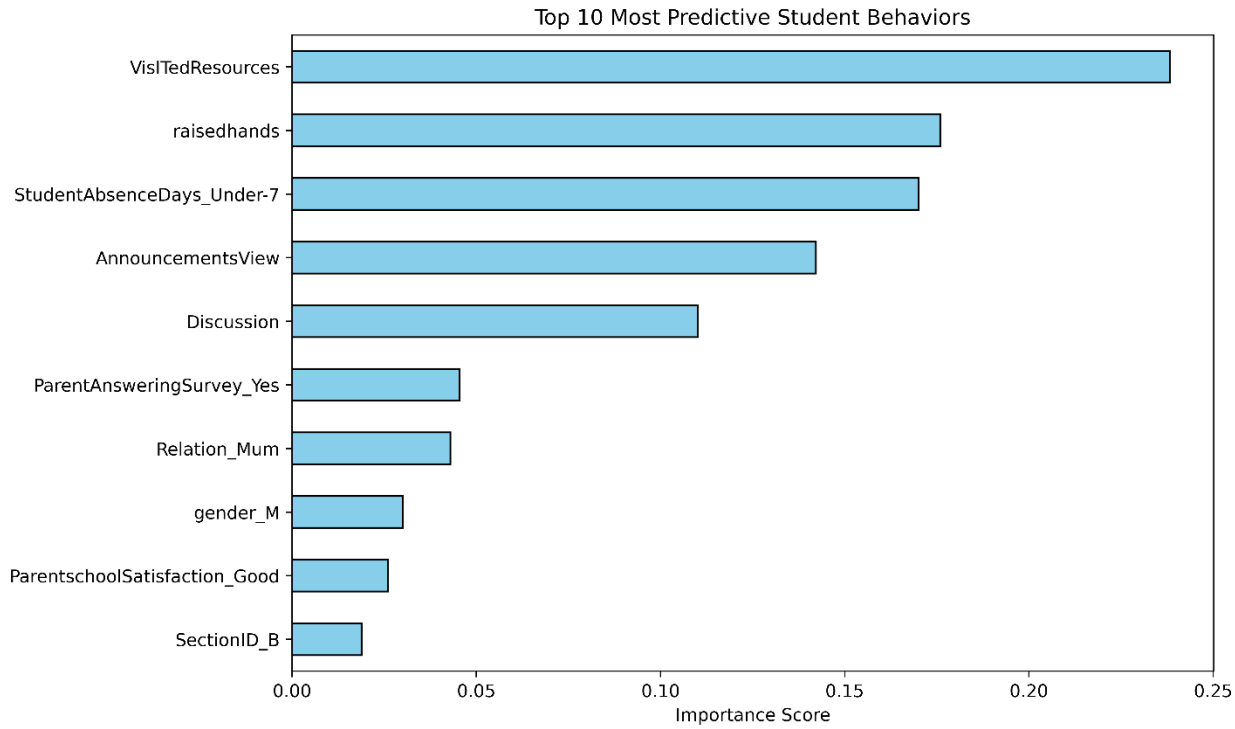
To improve model efficiency and reduce the risk of overfitting, feature selection techniques were applied. The final model utilizes the **Top 10** most influential features, selected based on their contribution to predictive performance.

Model evaluation was conducted using 5-fold cross-validation, where the dataset was partitioned into five subsets. In each iteration, four subsets were used for training, while the remaining subset was used for testing. This process ensures robust evaluation and reduces variance in model performance.

The final cross-validation results yielded a low standard deviation of **0.02**, indicating high model stability and consistency across different data splits.

4.3 Feature Importance Analysis

***Figure 1**: Feature Importance Ranking (Top 10)*



The Random Forest model provides an inherent mechanism for evaluating feature importance based on each variable's contribution to prediction accuracy.

*As illustrated in Figure 1, **behavioral engagement features** emerged as the most significant predictors of academic performance. In particular, VisITedResources and raisedhands were identified as the top contributors, collectively accounting for over 40% of the total feature importance.*

*In contrast, demographic variables such as gender and SectionID exhibited relatively low importance scores. This finding reinforces the hypothesis that student behavior and engagement patterns are more predictive of academic success than static **demographic characteristics**.*

These results highlight the practical value of focusing on actionable educational metrics, enabling educators to design targeted interventions based on student activity rather than fixed attributes

4.4 Model Selection

The Random Forest classifier was selected as the primary predictive model due to its robustness and strong performance in handling structured (tabular) data.

This algorithm offers several advantages:

- High resistance to overfitting through ensemble learning
- Ability to capture nonlinear relationships between features
- Strong performance on datasets with mixed feature types

- Built-in capability for feature importance estimation

Random Forest operates by constructing multiple decision trees during training and aggregating their predictions through majority voting. This ensemble approach enhances model generalization and improves predictive accuracy compared to single-tree models.

4.5 Hyperparameter Optimization

To enhance model performance, hyperparameter tuning was conducted using GridSearchCV in combination with feature selection.

The objective was to identify the optimal parameter configuration that maximizes predictive accuracy while maintaining model stability.

The best-performing configuration was:

Parameter	Value
n_estimators	300
max_depth	20
Criterion	entropy

This optimized configuration achieved a balance between model complexity and generalization, reducing overfitting while preserving high predictive power.

5. Results and Discussion

5.1 Model Performance

The final optimized model achieved the following performance metrics:

- Testing Accuracy: 86.46%
- Cross-Validation Accuracy: 71.25%

These results represent a significant improvement over the baseline model:

- Baseline Accuracy: 82.29%
- Baseline Cross-Validation Accuracy: 65%

This improvement demonstrates the effectiveness of feature selection and hyperparameter tuning in enhancing model performance.

***Figure 2**: Performance comparison between baseline and optimized models.*

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=====================================================================
              PROJECT PERFORMANCE SUMMARY TABLE
=====================================================================
                     Model Version  Feature Count Test Accuracy CV Accuracy(Avg) CV Std Dev             Status
          Initial Model (Baseline)             62        82.29%           65.00%      0.070        Overfitting
          Tuned Model (GridSearch)             62        82.29%           67.08%      0.055          Improving
Final Champion (Feature Selection)             10        86.46%           71.25%      0.020 Optimized & Stable
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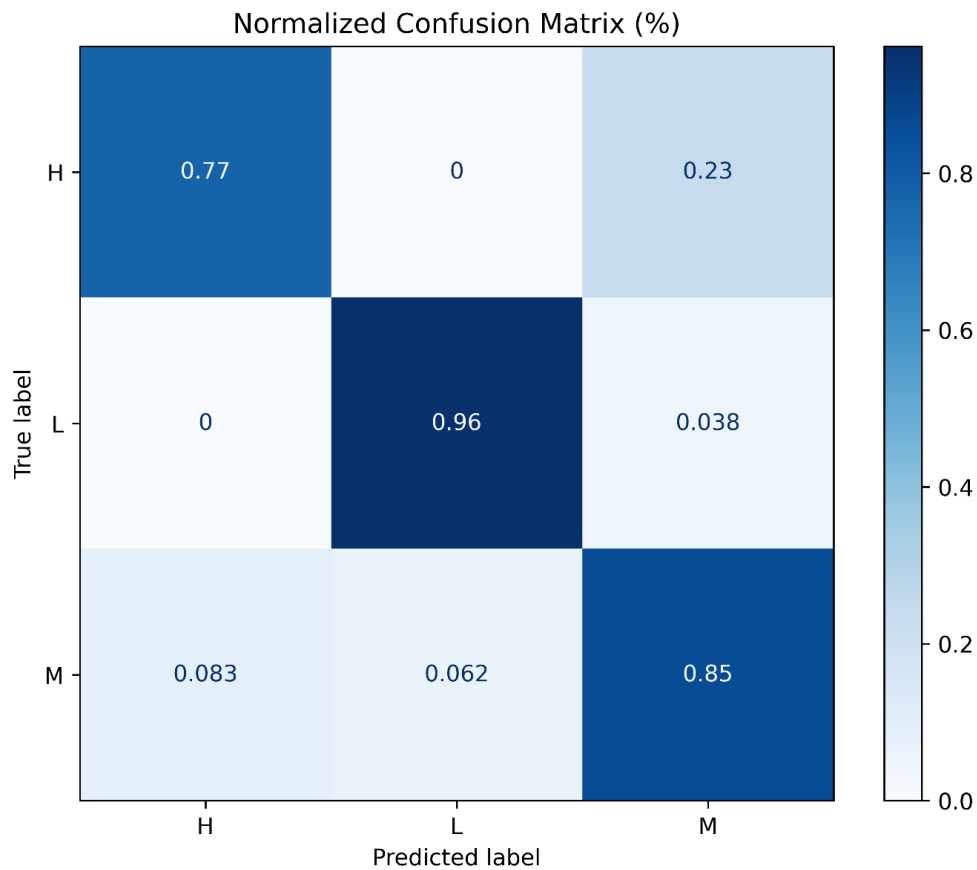
5.2 Confusion Matrix Analysis

Figure 3: Normalized Confusion Matrix of the Final Predictive Model



The normalized confusion matrix provides deeper insight into the classification performance across different student categories.

The model demonstrates:

- High recall (96%) for identifying *Low-performing (At-Risk)* students
- Zero misclassification between *High* and *Low* categories
- Moderate confusion between *Medium* and *High* classes (~23%)

The overlap between Medium and High categories can be attributed to similar behavioral engagement patterns, making them inherently more difficult to distinguish.

Importantly, the absence of confusion between High and Low classes (*0%)* highlights the model's reliability as an early warning system, ensuring that critically at-risk students are not misclassified as high-performing.

5.3 Feature Importance and Explainability

To enhance interpretability, SHAP (Shapley Additive Explanations) was used to analyze feature contributions.

The most influential predictors identified include:

- Student Absence Days
- Resource Engagement (VisITedResources)
- Class Participation (raisedhands)

These behavioral features act as key decision boundaries, particularly in distinguishing between Medium and High-performing students.

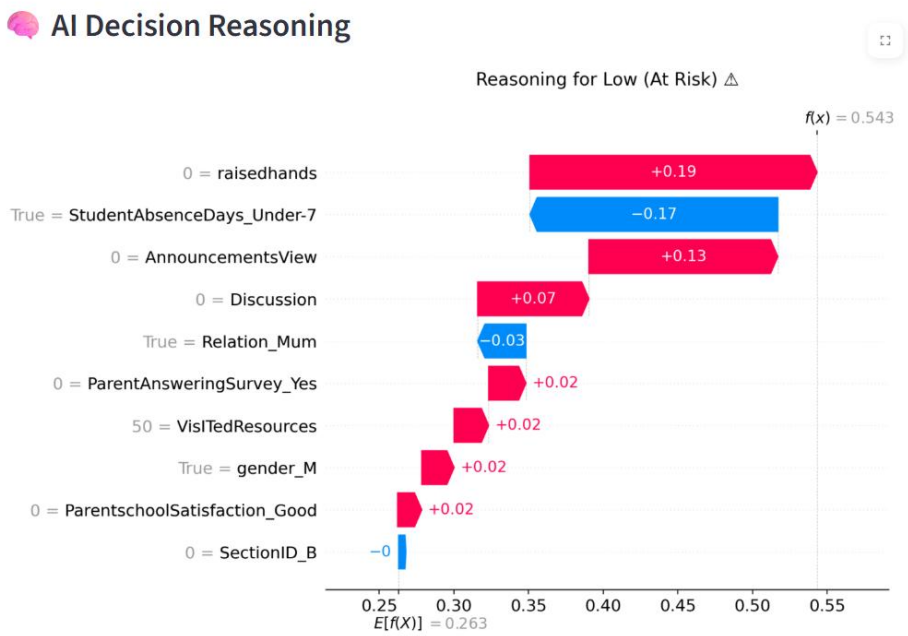
The findings reinforce the importance of student engagement metrics over static demographic characteristics.

5.4 SHAP-Based Case Analysis

To further interpret model predictions, SHAP waterfall visualizations were used to analyze individual student cases.

Case 1: Low-Performing Student ("At-Risk")

Figure 4: SHAP Waterfall Visualization for Low-Performing Student



The model prediction increased from a baseline probability of **0.263** to **0.543**, indicating a high likelihood of poor academic performance.

Key Risk Factors:

- Low class participation (*raisedhands = 0*)
- No interaction with announcements (*AnnouncementsView = 0*)
- No discussion activity (*Discussion = 0*)

These factors significantly increased the predicted risk.

Protective Factor:

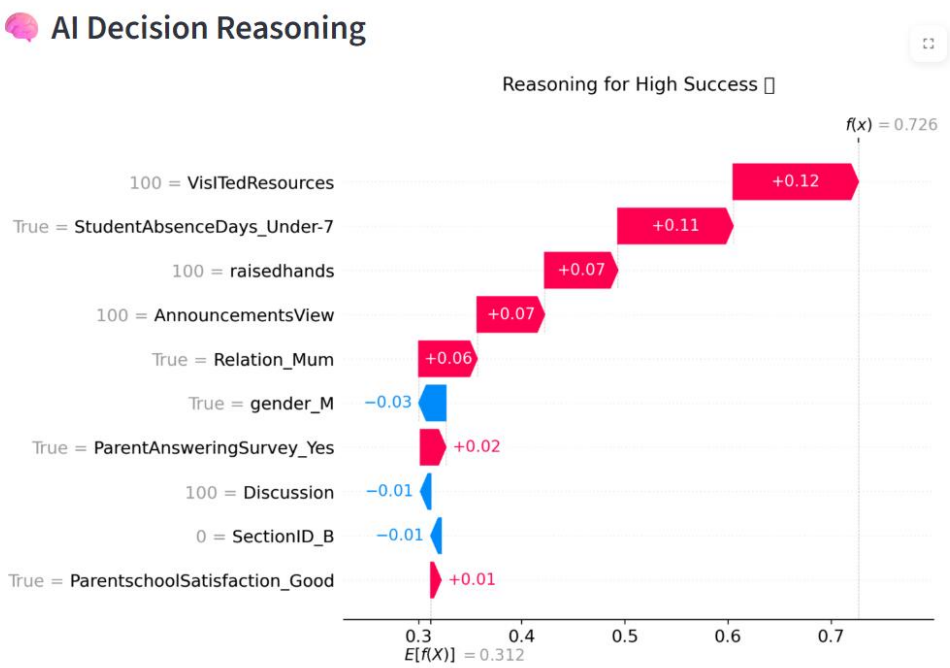
- Regular attendance (*StudentAbsenceDays < 7*)

Although attendance reduced the risk, it was insufficient to offset the negative impact of low engagement.

This case highlights the presence of "hidden at-risk students", learners who attend classes but remain academically disengaged.

Case 2: High-Performing Student

Figure 5: SHAP Waterfall Visualization for High-Performing Student



The model prediction increased from a baseline of **0.312** to **0.726**, indicating strong academic performance.

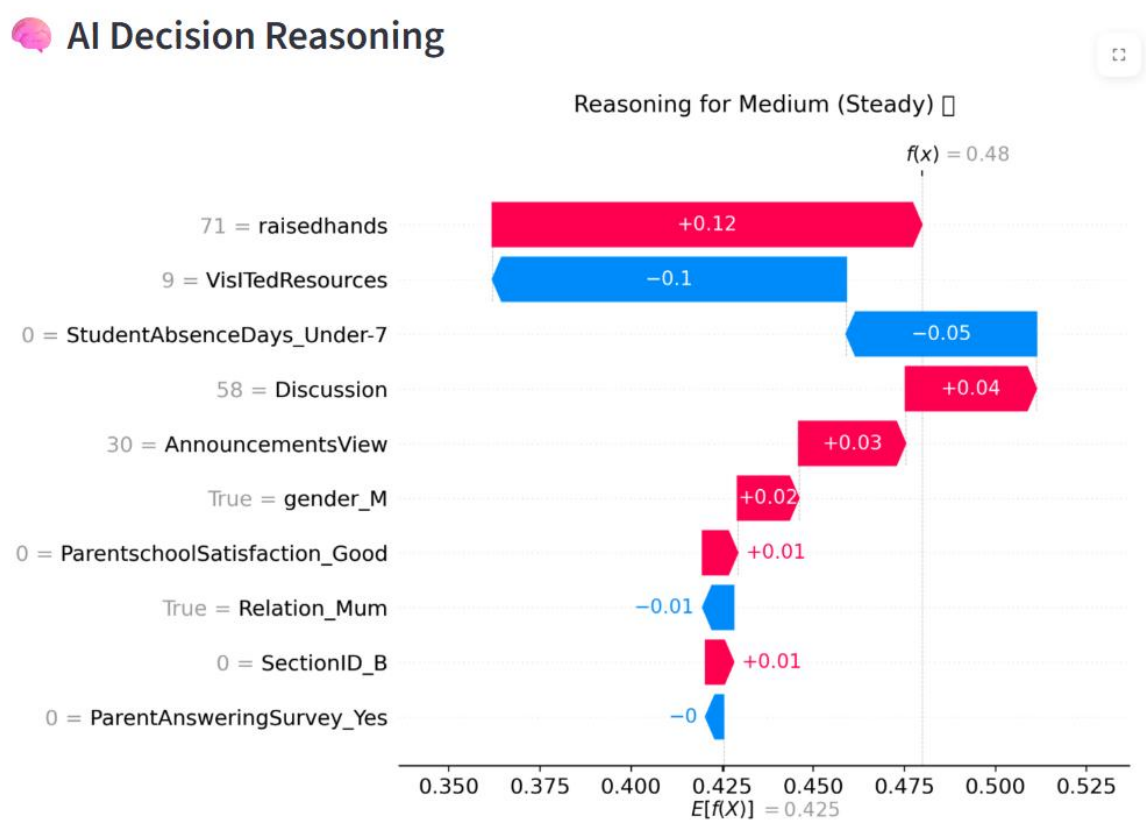
This outcome was primarily driven by:

- High resource engagement
- Active participation
- Consistent attendance
- Regular interaction with announcements

These behavioral patterns collectively contributed to a strong positive prediction.

Case 3: Medium-Performing Student (Mixed Engagement)

Figure 6: SHAP Waterfall Visualization for Medium-Performing Student



This case illustrates the model's ability to handle conflicting behavioral signals.

- Strong classroom participation (*raisedhands = 71*) contributed positively
- Low digital engagement (*VisITedResources = 9*) negatively impacted the prediction

The resulting probability (~48%) reflects a balanced classification, placing the student in the Medium category.

This demonstrates that the model does not rely on a single feature but evaluates multiple dimensions of engagement before making a decision.

5.5 Discussion

The results confirm that behavioral engagement metrics are the most significant predictors of academic performance.

The model successfully:

- Identifies at-risk students with high reliability
- Differentiates between performance levels using engagement patterns
- Provides interpretable insights through SHAP analysis

These findings support the use of machine learning based decision-support systems in education, enabling proactive and data-driven intervention strategies.

6. Conclusion

This study demonstrates that machine learning techniques can effectively predict student academic performance using behavioral engagement data derived from Learning Management Systems.

The findings highlight that academic success is not determined by a single factor, but rather by a dynamic interaction between engagement behaviors and absenteeism patterns. In particular, behavioral indicators such as *resource engagement* and *class participation* were found to significantly outweigh traditional demographic variables in predictive importance.

The application of Shapley Additive Explanations provided further insight into model behavior, revealing that engagement-related features contribute approximately 70–80% more predictive influence than demographic attributes such as gender or nationality. This reinforces the importance of focusing on actionable student behaviors when designing educational interventions.

With a predictive accuracy of **86.46%**, the EduPredict AI system demonstrates strong potential as a decision-support tool for educators. By enabling early identification of at-risk students, the model supports a shift from reactive grading practices to proactive, data-driven intervention strategies, ultimately improving student success outcomes.

This work demonstrates that explainable machine learning can bridge the gap between predictive analytics and practical educational decision-making.

7. Future Work

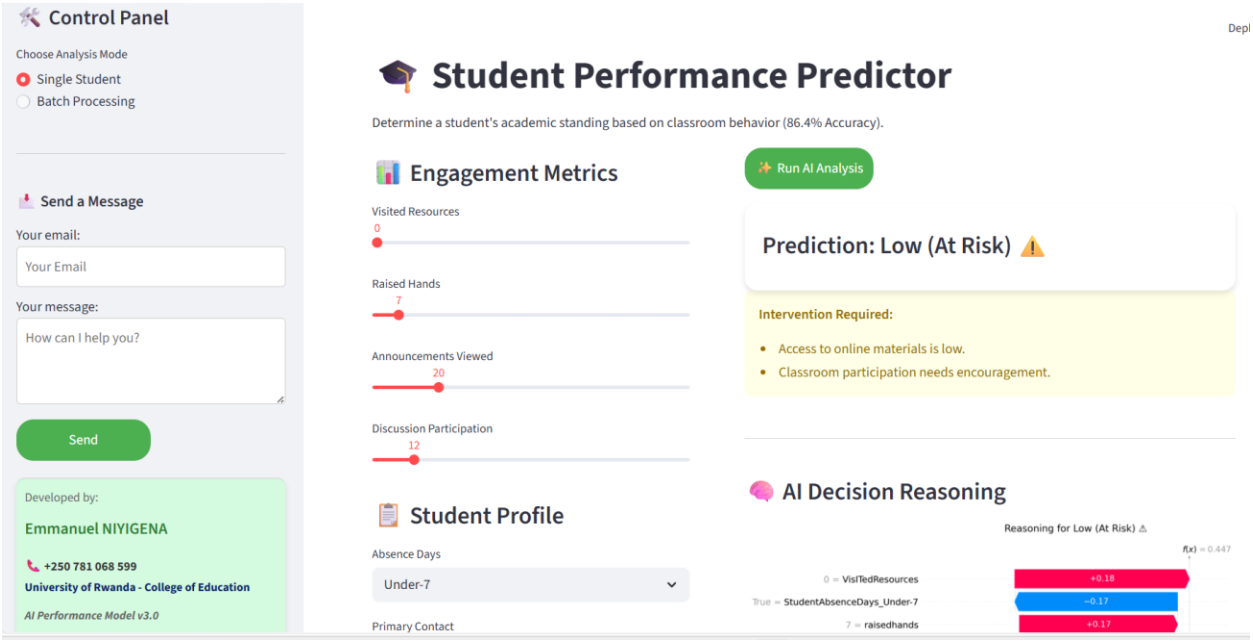
Future research directions may include:

- Integration of deep learning models to capture more complex behavioral patterns
- Incorporation of real-time LMS data streams for continuous prediction
- Deployment of the system within a live educational environment
- Expansion of the dataset to include more diverse educational contexts

These enhancements would enable real-time monitoring and adaptive intervention, further improving the effectiveness of predictive analytics in education.

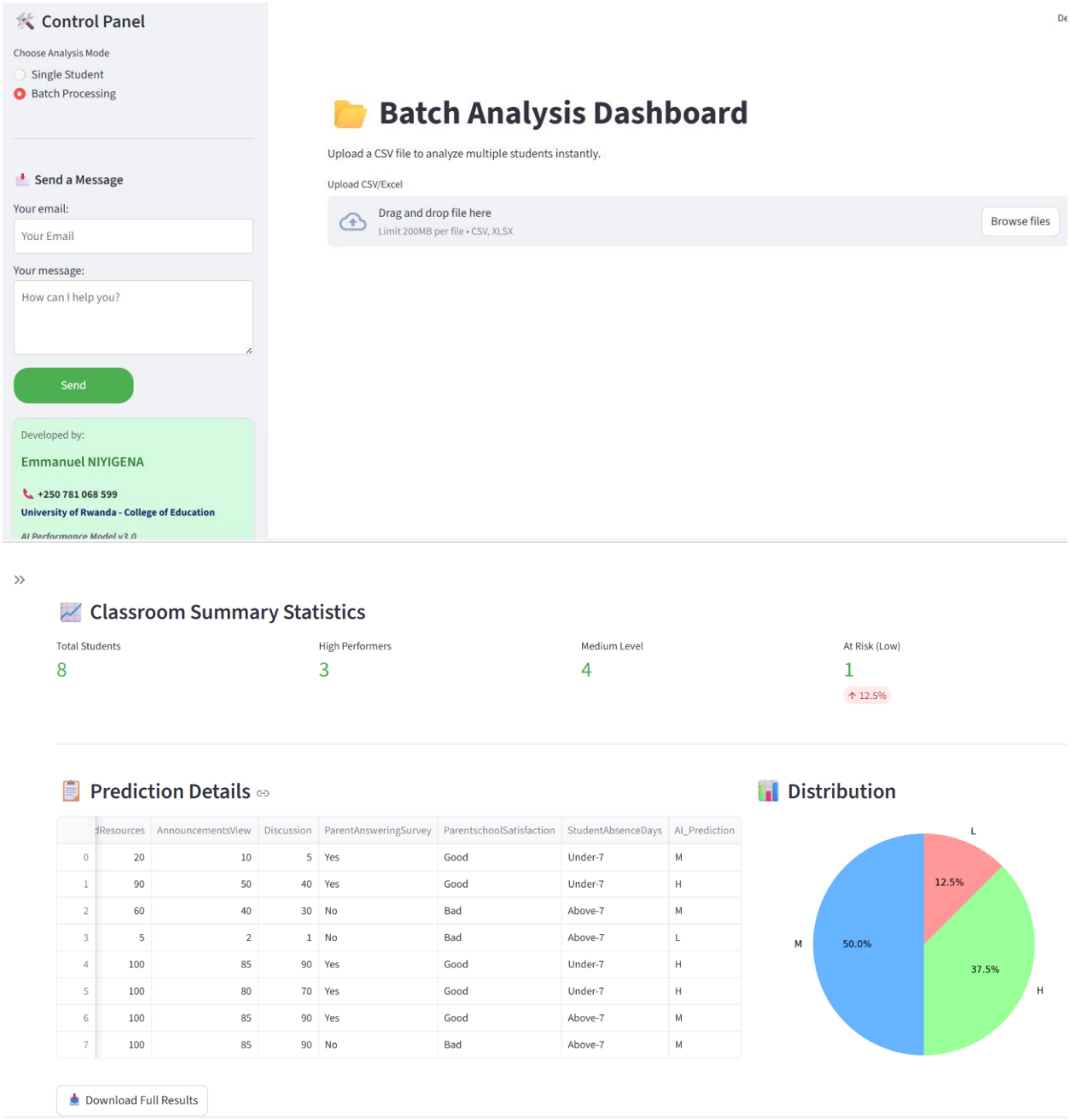
8. System Interface (UI) and Visualization

Figure 7: Prediction of a Single Student Using the EduPredict AI Interface



This interface allows educators to input individual student behavioral data and receive real-time predictions of academic performance. The system provides both classification results and interpretability insights, supporting informed decision-making.

Figure 8: Batch Prediction via CSV/Excel Upload



The system also supports batch processing, enabling educators to upload CSV or Excel files containing multiple student records. This functionality allows for scalable analysis of entire classrooms, facilitating efficient identification of at-risk groups.

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